

Fixing ThinPro Connection Startup Loop

<https://gemini.google.com/app/d17e61c67c48434c>

User prompt: Help, after creating the custom connection, it kept on opening a new application and looping

Response: This behavior (opening a new application instance over and over in a loop) happens because of how HP ThinPro's Connection Manager handles **custom executable scripts** combined with **autostart or reconnect rules**.

When ThinPro launches a Custom Connection, `adaemon` monitors the launched process ID (PID). Because `perspectiveworkstation.sh` is a shell script wrapper that launches a background Java/Chromium process and then exits, ThinPro thinks the application has crashed or closed. If **Autostart**, **Keep-Alive**, or **Automatic Reconnect** is enabled in ThinPro, `adaemon` immediately attempts to relaunch it, leading to an infinite loop.

Here is how to resolve the issue directly.

Step 1: Prevent ThinPro from Relaunching the Script

1. Open **ThinPro Control Center / Connections**.
2. Select your **Perspective Workstation** custom connection and click **Edit**.
3. Go to the **Advanced** or **Automatic Action** settings.
4. Ensure options like **"Keep Alive"**, **"Restart on Exit"**, or **"Reconnect automatically"** are **DISABLED**.
5. Keep **Autostart** set to 1 (or checked), but turn off any setting that forces process monitoring/re-execution upon exit.

Step 2: Use `exec` in the Shell Script (Fix Process Hand-off)

When `perspectiveworkstation.sh` runs, the shell script completes, but the actual GUI process branches off. Adding `exec` ensures the Java process replaces the shell process directly, so ThinPro tracks the actual running application rather than a temporary script.

1. Edit your script:

```
nano /persistent/ignition/perspectiveworkstation/perspectiveworkstation.sh
```

2. Find the final line in the script that actually launches Java or the Perspective executable (usually starting with `java ...` or `./app ...`).
3. Add `exec` to the front of that final execution command.

For example, change:

```
"$JAVA_EXEC" -jar launcher.jar
```

To:

```
exec "$JAVA_EXEC" -jar launcher.jar
```

4. Save and exit (Ctrl+O, Enter, Ctrl+X).

Step 3: Add a Single-Instance Lock Strategy

If ThinPro is triggering multiple triggers on startup, wrap the startup script with `flock` to prevent more than one instance from running at the same time:

1. Update the **Command line** in the ThinPro **Connection Manager** custom connection to:

```
flock -n /tmp/perspective.lock /persistent/ignition/perspectiveworkstation/perspectiveworkstation.sh
```

How this works: `flock -n` tries to acquire an exclusive lock file (`/tmp/perspective.lock`). If an instance is already running or launching, any secondary attempt spawned by ThinPro will instantly fail and exit cleanly instead of opening another window.

Verification

How to verify the fix:

- Close Perspective Workstation manually or reboot the thin client.
- Only **one** window should open on boot.
- Closing the application should safely return you to the desktop without triggering an endless respawn loop.

Response: Looking at the image of `/persistent/ignition/perspectiveworkstation/perspectiveworkstation.sh`, the exact line responsible for the looping issue is in the `# Launch` section at the very bottom.

Currently, the command runs `"real_dir"/runtime/bin/java` as a child process of the shell script. Because it isn't prepended with `exec`, ThinPro's `adaemon` service tracks the parent Bash process, loses track when the wrapper returns/forks, and repeatedly re-spawns it.

Here is how to modify the script directly:

Step 1: Open the Script for Editing

In your terminal on the thin client, open the script using `nano`:

```
nano /persistent/ignition/perspectiveworkstation/perspectiveworkstation.sh
```

Step 2: Edit the # Launch Section

Scroll down to the bottom section labeled `# Launch`. It currently looks like this:

```
# Launch
APP_DIR="$real_dir" "$real_dir/runtime/bin/java" \
--add-exports=java.desktop/sun.awt=ALL-UNNAMED \
--add-opens=java.desktop/java.awt=ALL-UNNAMED \
-Dsun.java2d.uiScale="$gdk_scale" \
-jar "$real_dir/app/$jar_name" "$@" /dev/null 2>&1 &
```

Modify that section to look like this:

1. Add `exec` right after `APP_DIR="$real_dir"`.
2. Remove the trailing `&` at the end of the last line (and remove the broken `/dev/null 2>&1` if you want clean process handoff).

Updated Code:

```
# Launch
APP_DIR="$real_dir" exec "$real_dir/runtime/bin/java" \
--add-exports=java.desktop/sun.awt=ALL-UNNAMED \
--add-opens=java.desktop/java.awt=ALL-UNNAMED \
-Dsun.java2d.uiScale="$gdk_scale" \
-jar "$real_dir/app/$jar_name" "$@"
```

Step 3: Save and Test

1. Press `Ctrl + O` then `Enter` to save the file.
2. Press `Ctrl + X` to exit `nano`.
3. Test launching the Connection from ThinPro's Connection Manager or reboot the machine.

Why this fixes it: By adding `exec` and removing the backgrounding ampersand `&`, Java replaces the shell process directly in memory instead of spawning as an unmonitored background job. ThinPro's connection manager (`adaemon`) now directly monitors the main Java window process ID, preventing it from falsely registering an immediate process exit and looping.

How to verify:

- Launch the connection; it should open exactly one instance of Perspective Workstation.
- Check running processes with `ps aux | grep java` to confirm only one process is active.